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I want to begin with a question: where am I standing when I look at a painting? My project began with Velázquez's *Las Meninas*, but it eventually connected a seventeenth-century room, a worn charging cable, my own drawing, a small convolutional autoencoder, and a four-second moving image. The common thread is the position of the viewer—how a work of art can make that position visible, and how a neural network reveals a very different kind of looking.

2

I chose *Las Meninas* because it disrupted the fast, classificatory way I had learned to look at art. As a child, I entered pictures slowly through books and illustrations. Later I began to identify the subject, period, and style, then move on. *Las Meninas* resisted that habit. The painter, the enormous canvas, the mirror, the rear doorway, and several outward gazes all make my own position part of the composition. I was no longer only looking at a painting; the painting seemed to organize the place from which I looked.

3

My ready-made is called *Tether*. It is an ordinary nylon-braided charging cable that came with the Mac I began using at university. Within one year, both connectors had frayed. When I was younger, I could not understand why my mother's cables broke while mine stayed new. Now the positions have reversed because so much of my reading, work, and attention happens through a screen. By selecting and naming the cable, I treated its damage as evidence: the object had recorded my habits. Like a ready-made, its meaning came from recontextualization rather than fabrication.

4

I turned the ready-made into a drawing called *The Tethered Viewer*. The room borrows the perspective structure of *Las Meninas*, but I placed my own hands and the damaged connector in the foreground. The red cable runs toward the mirror and doorway. It becomes more than an object: it is a line that joins childhood to university, physical wear to digital dependence, and the viewer's body to the architecture of the painting. The drawing makes visible the position that *Las Meninas* made me feel.

5

For the neural experiment I intentionally used a small convolutional autoencoder rather than a very large model. It saw my drawing at 128 by 128 pixels. I trained it on 800 synthetic images containing perspective lines, nested rectangles, grids, arcs, radial connections, and occasional red paths. It never saw paintings, faces, or handwriting. With only 27,983 parameters, the network learned a limited geometric vocabulary. This makes the results easier to interpret: whatever survives should reflect recurring structure, not knowledge of *Las Meninas*.

6

This is the first convolutional stage. The input is transformed into twelve feature maps at 64 by 64 pixels. Different channels respond strongly to different local patterns—edges, diagonals, repeated frames, and color contrast. This resembles the course discussion of receptive fields: each filter combines information from a small neighborhood and produces a weighted response. But it is important not to anthropomorphize the maps. There is no “mirror channel” or “cable channel.” The network is separating visual tendencies, not recognizing the personal meaning of the drawing.

7

The bottleneck is the narrowest representation in the network: thirty-two channels at 16 by 16 pixels. The original input contains 49,152 values; the bottleneck contains 8,192. The network must decide what is useful enough to carry forward. Coarse relationships survive—the room frame, openings, diagonals, and approximate figures. Handwriting, faces, and the exact shape of the red tether do not. This is the most important idea in the experiment. The bottleneck is not an interpretation of my story. It is a constrained geometric summary. In a broader cognitive sense, every system of seeing preserves some relationships and discards others.

8

Here is one actual forward pass from left to right. The encoder reduces spatial resolution from 128 to 64, then 32, then 16 pixels. The decoder expands the bottleneck back to the original size. Notice that expansion is not the same as recovery. The room frame, central rectangle, major directions, and approximate figures reappear, but handwriting and faces remain blurred. The red tether becomes weaker and more diffuse because the network has learned a broad prior for geometric red paths, not this particular autobiographical line.

9

The final image returns to *Las Meninas* through a style-transfer workflow. The content image supplies the figures and room; the project-generated cognitive-map image supplies layered planes, nodes, lines, and a stronger sense of convergence. In the final composition, the painter, large canvas, mirror, doorway, Infanta, attendants, and dog remain recognizable, while the geometry of looking becomes visible on the surface. I also inserted a contemporary viewer into the mirrored position. The transformation is not meant to imitate Velázquez. It makes the relation between room, painting, and viewer more explicit.

10

I ended with a four-second looping parallax study. The code assigns a simple vertical depth proxy, so lower and slightly more central regions move more strongly than distant regions. The virtual camera follows a smooth sine-and-cosine loop, and each frame remaps the image with reflected borders. This is a heuristic 2.5D effect, not CNN depth estimation. It feels like entering the picture because differential motion separates the image into depth layers: foreground figures drift, while the central doorway remains comparatively stable and acts like a visual anchor. I want to end with one sentence: every way of seeing has a bottleneck. The network preserves geometry; I preserve the history and meaning attached to the line.